

CareerAtlas

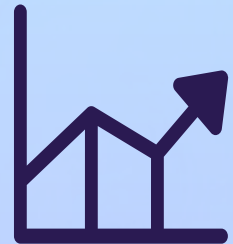
CIS 2450 Final Project

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Moving to a New City Shouldn't be a Leap of Faith

- Salary ≠ quality of life
- Data is scattered and hard to compare
- No career-specific guidance by city
- Anecdotal Evidence Isn't Enough



CareerAtlas Removes the Guesswork

DATA ALL IN ONE PLACE

Combines salary, rent, and job data into a single view. Eliminates the need to cross-reference multiple sources.

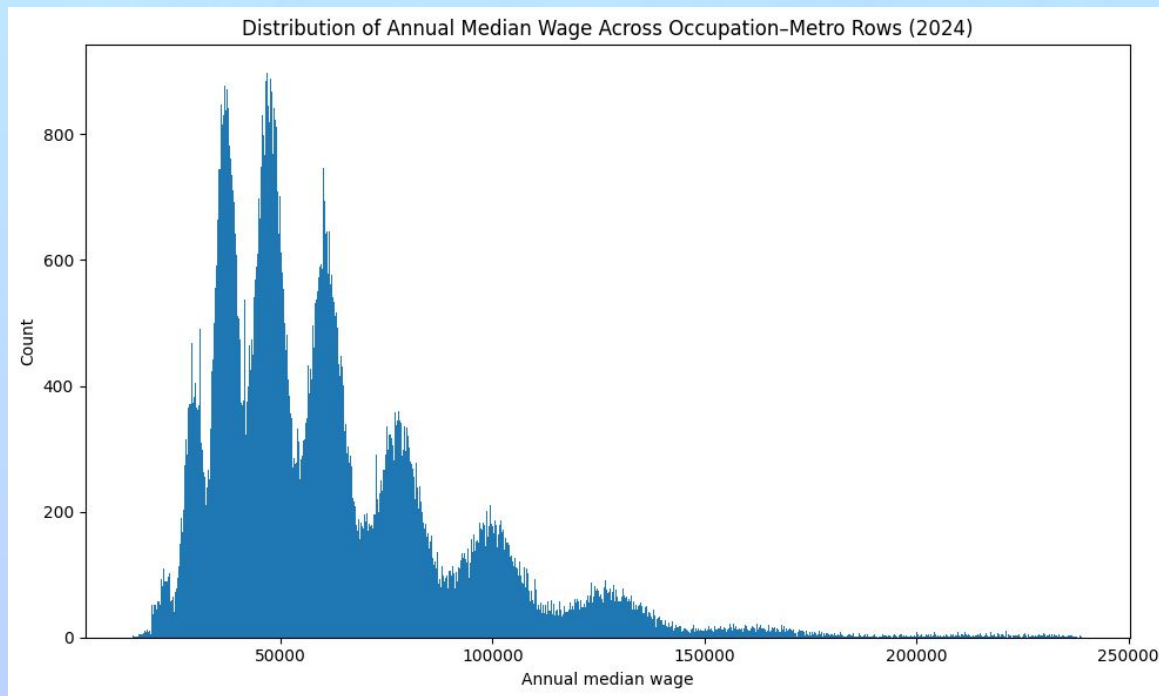
CAREER SPECIFIC INSIGHTS

Tailors recommendations based on your specific role. Highlights how outcomes vary within the same city across jobs. Reveals tradeoffs between salary, cost, and opportunity.

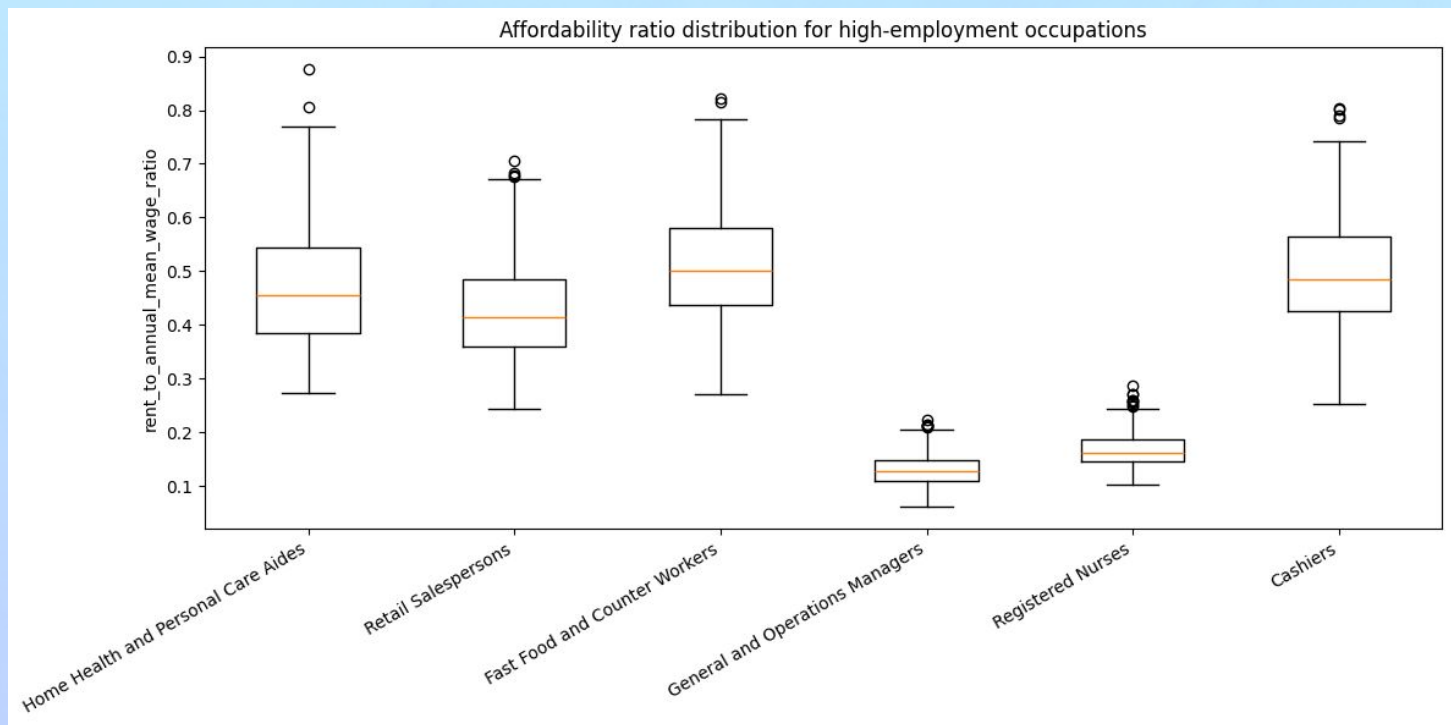
PREDICTIVE CAPABILITIES

Uses modeling to estimate affordability across cities. Captures complex relationships beyond simple averages.

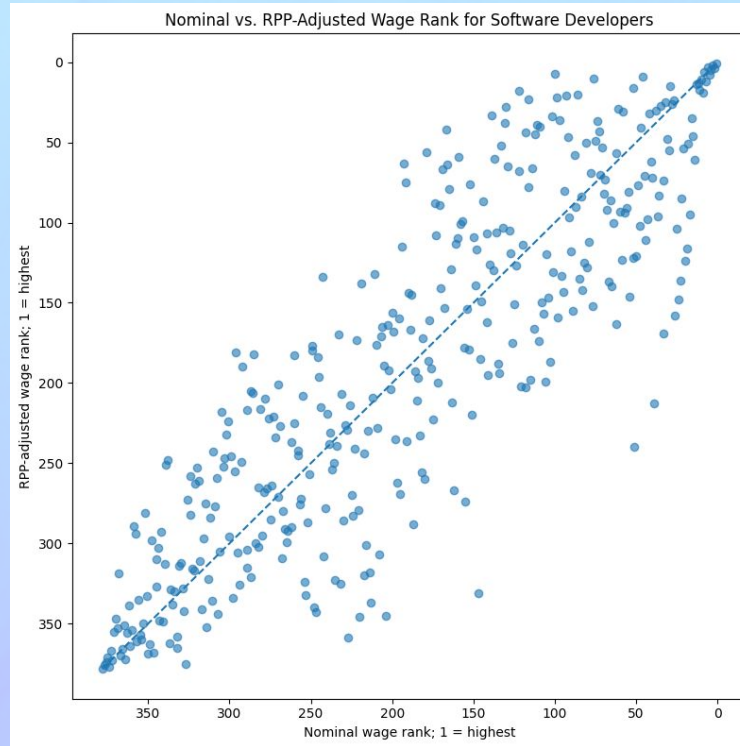
EDA Findings



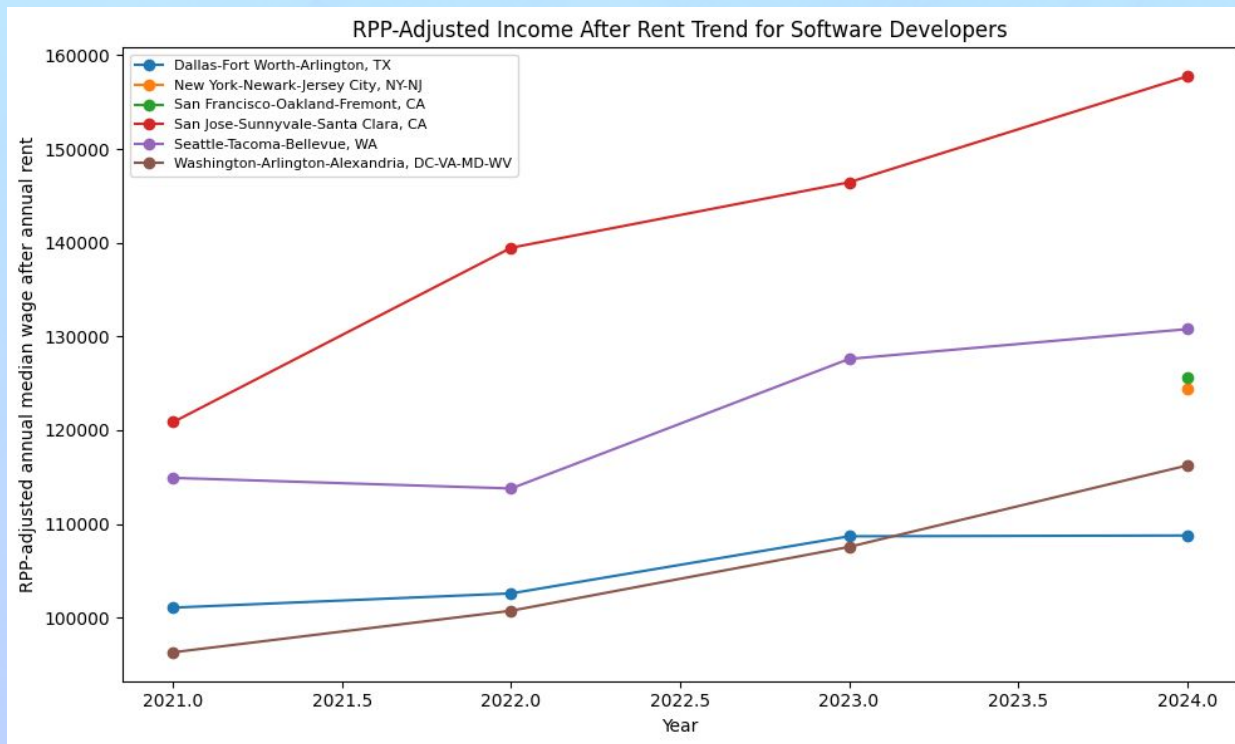
EDA Findings



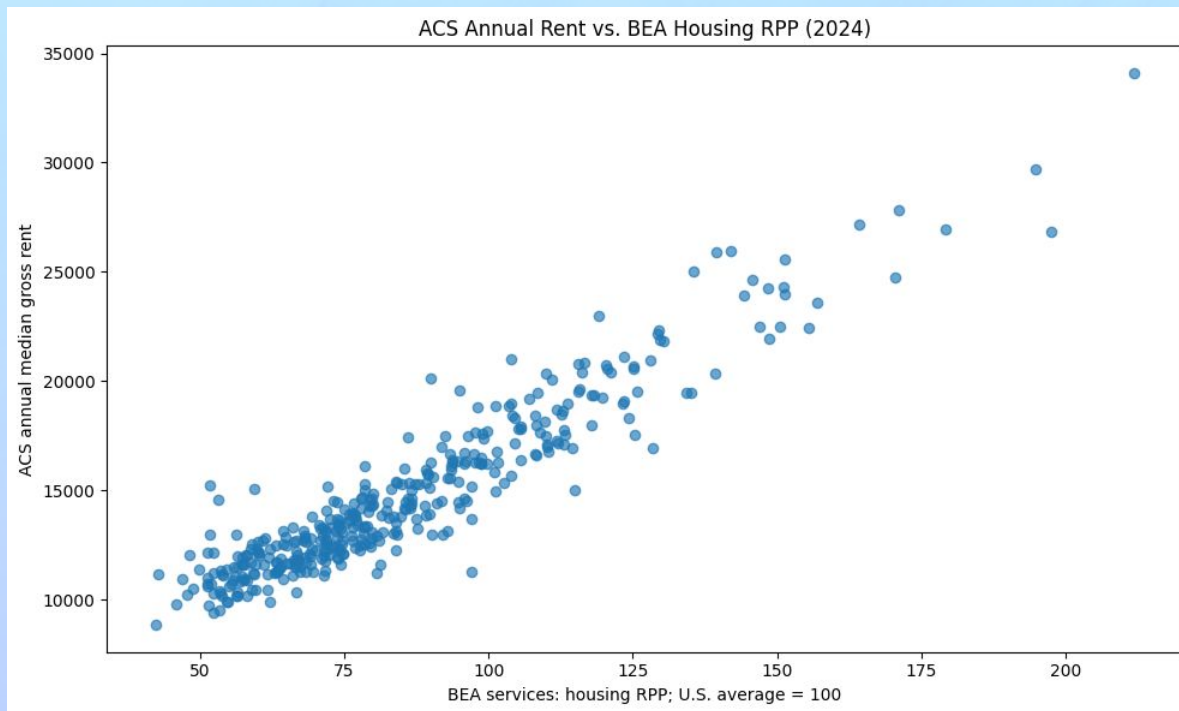
EDA Findings



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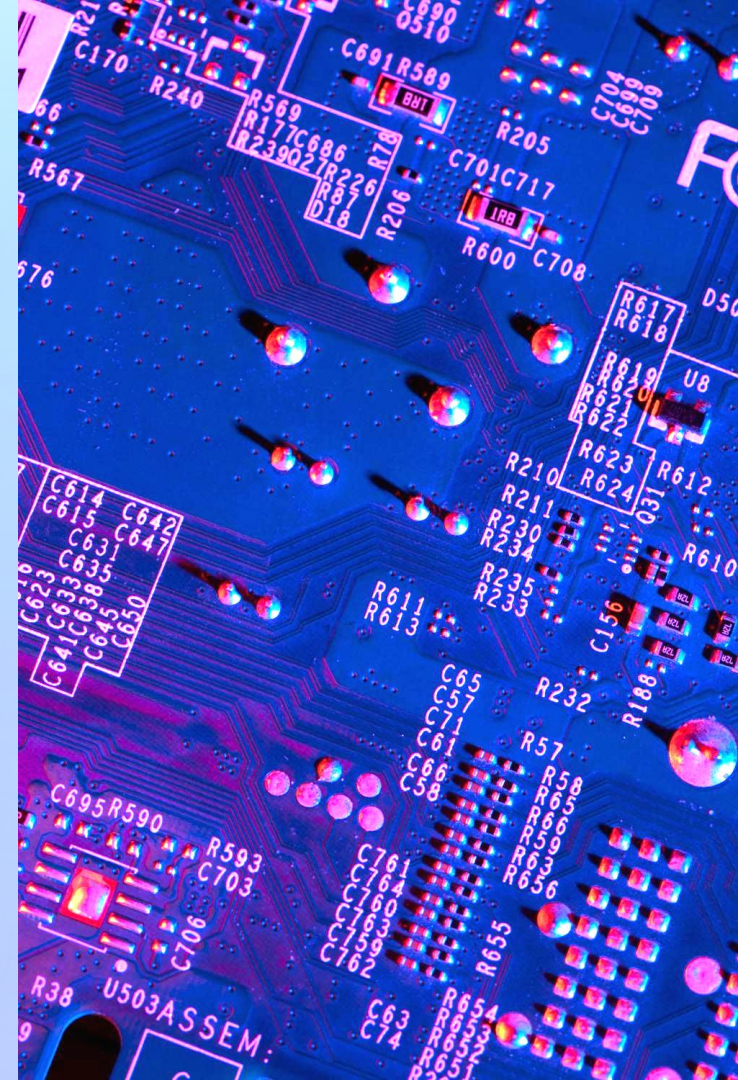


EDA Findings

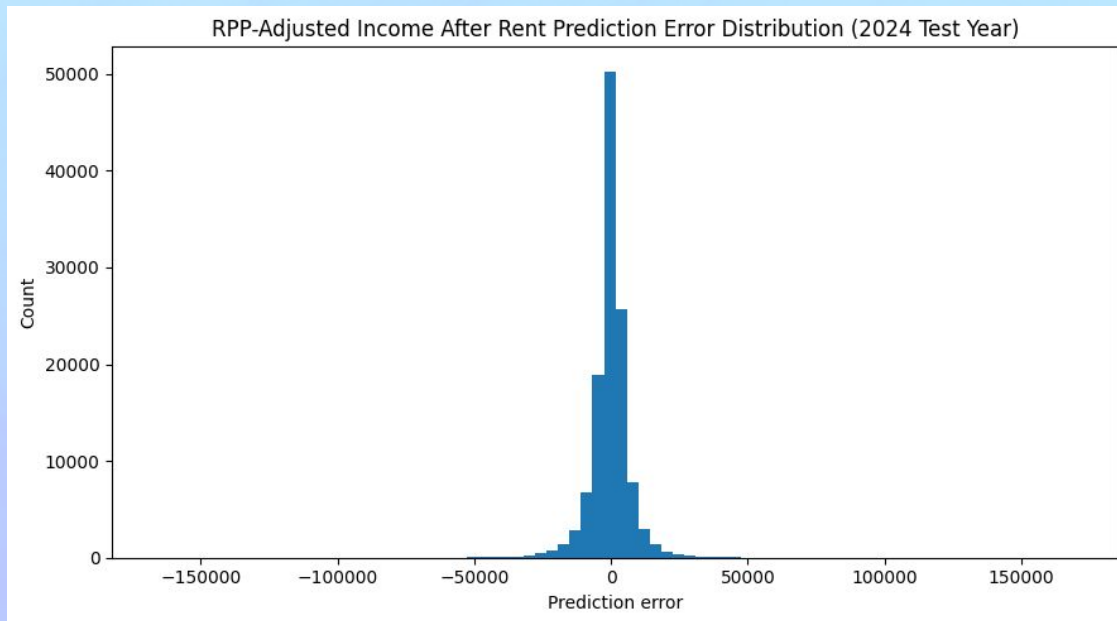
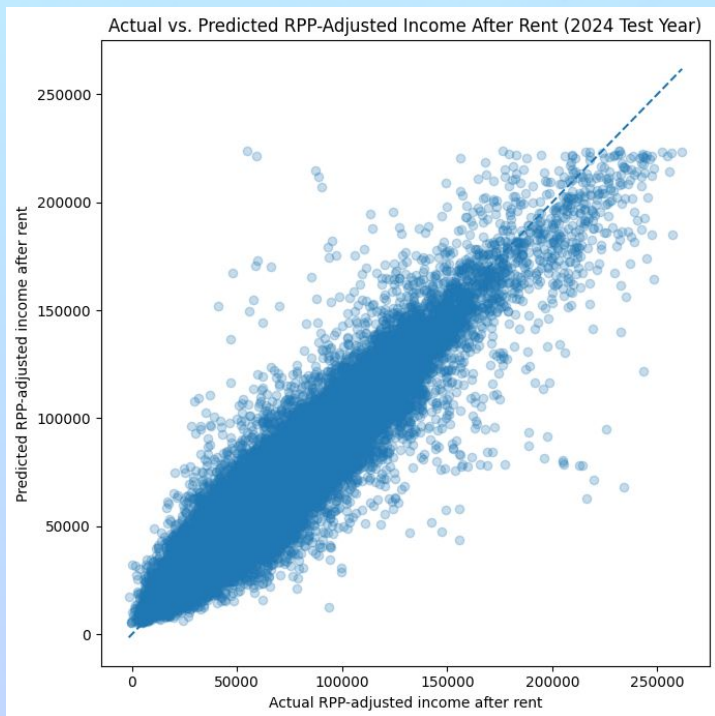


The Models

- Built **career-city scoring model** using:
 - RPP-adjusted wages (cost of living)
 - Rent affordability
 - Employment + job concentration
- Created **metro-level features** by aggregating data across occupations
- Applied **K-Means clustering** to group similar cities
 - Used features like wages, rent, population, and cost of living
 - Visualized clusters with PCA
- Developed **nearest-neighbor model**
 - Finds cities most similar to a given city
- Tested **forecasting models** (e.g., regression, random forest)
 - Predict future trends using time-based splits



The Models





Implications and Insights

- Don't fall for the “high salary” trap
- Small Cities are underrated
- Career choice matters as much as location

LIMITATIONS

Upward Mobility isn't Factored in

- Analysis reflects **current salaries only**, not long-term growth
- Some cities may offer **better career progression over time**

Buying power isn't the same as quality of life

- Rent-to-wage ratio captures affordability, not lifestyle factors
- Excludes things like taxes, safety, transportation, and personal fit

THANK YOU

